

## CHAPTER 11

# Human Factors

### 11.1. Perspective

In designing productive work systems, I.E.s put together various components that will operate in such a way as to achieve the outputs desired from the system. The components making up a productive work system include:

- **Equipment**—Machines, material handling devices, gauges, instruments, etc.
- **Processes**—Chemical treatments, thermal treatments, laser hardening, etc.
- **Materials**—Raw materials, purchased parts, lubricants, chemicals, supplies, etc.
- **Capital**—Cash, credits, debits, investments, etc.
- **Energy**—Electrical, steam, mechanical.
- **Facilities**—Buildings, other structures.
- **Information**—Forecasts, sales records, costs, production orders, vendor histories, shop floor status, etc.
- **People**—Workers, supervisors, design engineers, quality control engineers, salespeople, managers, etc.

When the I.E. specifies a specific machine to include in the system, he does so knowing the performance characteristics of that machine. Among the characteristics known about a typical machine are:

- *Range of Operations*—Operations it is capable of performing.
- *Size of Workpiece*—The machine can accommodate a workpiece up to a certain size and shape.
- *Speed of Operation*—Spindle speed, etc.
- *Tolerance Capabilities*—The tolerances the machine is capable of holding.
- *Energy Consumption*—Energy consumed in operation.
- *Maintenance Requirements*—Frequency of overhaul, other maintenance operations.

When the I.E. specifies that a human being should be included as a specific component in the system, it is equally important that proper consideration be given to the performance characteristics of that person. People have performance characteristics just as machines do.

Figure 11.1 illustrates some of the performance characteristics of a "typical" human being compared to a "typical" machine tool. It should be clear from these comparisons that human beings are ideally suited for many functions in a system, but woefully inadequate for others.

The field of study that is concerned with quantifying and measuring the performance characteristics of human beings is called *human factors engineering* or simply *human factors*. Most of the research and teaching associated with human factors is conducted within industrial engineering departments, usually at the graduate level. Human factors is an interdisciplinary field that includes professionals from the behavioral sciences (e.g., psychologists and sociologists) and the biological sciences (e.g., physiologists and bioengineers) as well as industrial engineers.

Human performance characteristics fall into two broad categories, physiological and psychological. *Physiological performance characteristics* have to do with the physical aspects of human activities (we are concerned primarily with work activities), such as lifting, reaching, carrying, hearing, seeing, and speaking. *Psychological performance characteristics* are those dealing with the mental aspects of human activities, such as stress, boredom, and motivation.

## 11.2. Physiological Aspects of Human Performance<sup>1</sup>

Human beings, by virtue of the biological characteristics of their bodies, are limited in what they can do in a work system. Specifically, the abilities of people to perform various types of motor activities depend on the physical structure of the skeleton, the skeletal muscles, the nervous system, and the metabolic processes.

We are concerned primarily with the bones comprising the upper and lower extremities and the bones of the spine, since these are concerned with the execution

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<sup>1</sup>Our coverage of human factors will necessarily be brief. For more complete coverage, see, for example, M. S. Sanders and E. J. McCormick, *Human Factors in Engineering and Design*, 6th ed. (New York: McGraw-Hill Book Company, 1987).

| <i>Performance Characteristic</i>  | <i>"Typical" Machine Tool</i>                                                                                      | <i>"Typical" Human Being</i>                                                                                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of Operations                | Most M.T.'s perform one basic operation; some perform up to 10.                                                    | Extremely broad range.                                                                                                                                                            |
| Size of Workpiece                  | Some machines perform operations on microscopic-sized workpieces; others machine workpieces weighing several tons. | Very limited range. Cannot perform operations on extremely small items. Can perform repetitive operations on items weighing up to 50 pounds, where physical handling is required. |
| Speed of Operation                 | Can be extremely fast.                                                                                             | Usually very slow.                                                                                                                                                                |
| Tolerance Capabilities             | Machines routinely hold tolerances to 0.001 inch.                                                                  | Very poor at accuracy and repeatability.                                                                                                                                          |
| Energy Consumption                 | Very efficient in converting energy into useful output.                                                            | Very low level of energy output; cannot sustain output on even level.                                                                                                             |
| Maintenance Requirements           | Regular maintenance required; complete overhaul needed periodically.                                               | Needs minor "maintenance" frequently, such as water, food, oxygen, rest room breaks, band-aids, etc. Major overhauls are performed continuously, as the body mends itself.        |
| Response to Unexpected Occurrences | Very limited capability.                                                                                           | Extremely resourceful and creative.                                                                                                                                               |

**Fig. 11.1.** Performance characteristics of a typical machine tool vs. a human being.



of physical activities. These skeletal bones are connected at body joints, such as the fingers, wrists, elbows, shoulders, hips, knees, ankles, and feet. The manner in which they are shaped and connected puts limitations on our ability to perform physical activities. The wrist will rotate only so far. Our knees bend in one direction but not the other. Our thumbs oppose the four fingers, giving us considerably more dexterity in our hands than any other animal. (Some anthropologists claim that this characteristic is at the root of human mental superiority in the animal world.)

The bones are held together at their joints by ligaments. The skeletal muscles are connected to different bones through tendons, such that when the muscles are activated, the bones apply mechanical leverage. The fibers of the muscles convert chemical energy (from food) into mechanical work, such as lifting, walking, and so on.

The actions of the muscles are controlled by motor nerves. The nerves convey conscious and subconscious messages from the brain to the muscles, causing physical activities to occur. A machine operator reads a work instruction from a blueprint, then executes that instruction by sending the appropriate message over the nervous system to the appropriate muscles, which move bones so as to accomplish the specified task.

Muscles contract to move bones. Contraction requires energy. As it is consumed, it must be replenished through the circulatory system. The heart continuously pumps blood through the circulatory system. The blood picks up nutrients from the digestive process and it replenishes its supply of oxygen from the lungs. The nutrients and the oxygen are carried by the blood to the muscles throughout the body.

Researchers in human factors have carefully studied the process described above and have determined the energy requirements for various types of jobs. Based on these data, jobs can be designed which can be performed by a large percentage of people likely to be involved in such work. Also, optimal rest periods can be specified.

Substantial amounts of detailed information have been generated concerning the biomechanics of motion of the human body. For example, tables of data exist which characterize the movements of the arms, legs, and other body members. As a specific example, the average person's elbow can bend 142 degrees. People at the 5th percentile can bend their elbows only 126 degrees, while people at the 95th percentile can bend their elbows 158 degrees.

Data such as these are extremely useful in designing workplaces for optimum human performance. They are also very useful in designing tools. Better work methods can also be determined, such as having people lift with the large muscles of their legs rather than with the much weaker muscles of their back.

### 11.3. Psychological Aspects of Human Activities

The physiological aspects of human activities discussed in the preceding section view the human being as a mechanism, much as one would view a machine. While

the human body is, indeed, a mechanism governed by physical laws, it also contains a thinking, reasoning, feeling brain. Human beings experience pain, joy, sadness, depression, anger, boredom, frustration, fear, outrage, jealousy, love, hate, and (occasionally) schizophrenia.

We have not made very much progress in quantifying, measuring, or understanding the psychological aspects of human performance. We can predict with much greater accuracy the limit of how much a person can lift than the limit of how much boredom that same person can withstand.

Do people respond better in large open offices or smaller walled ones? With windows or without? Does soft music stimulate thinking, or slow down the work pace? Does fast music result in more output per hour? Unfortunately, the research that has been performed on these questions is inconclusive. Individual differences on such questions are great. Also, people tend to be very adaptable and seem to quickly become oblivious to many such factors.

We can only make some broad, sweeping statements about the psychological aspects of human activity:

- *People are still motivated by money.* Even among people in higher income brackets, the size of their salary is a quantitative measure of their relative worth to the organization.
- *People would rather be doing productive work than be idle.* This has less to do with the "work ethic" than with basic human nature. People are bored when they have nothing to do.
- *Many American workers do not necessarily want "job enrichment."* Behavioral scientists have long proclaimed that workers want more variety and more challenge in their work. Recent attempts to implement "job enrichment" programs have met with surprising resistance.
- *Workers themselves often have the best ideas on how to increase productivity, if they were only asked.* It is astounding that most American managers have not yet learned that their employees have brains as well as hands.
- *Most people respond positively when they know the objectives of their organization, what their own jobs are, and that they will be treated fairly, consistently, and with respect.*

## 11.4. Human Interface with the World of Work

An important dimension of human performance is the various ways in which human beings interact with their world. Some of the important interfaces are with the following entities:

- *Work Environment*—The physical environment in which the productive work system exists.
- *Social Structures*—Basic values, creeds, laws, traditions, community arrangements, legal structures, educational systems.



- *Machines*—The physical components of equipment and processes in the work place.
- *Information/Communication Systems*—The total set of systems and procedures by which work system components exchange information.
- *Organizational/Supervisory Structure*—The total set of authority/responsibility/accountability relationships within the work system.
- *Robots/"Intelligent" Machines*—The rapidly emerging set of electro-mechanical devices which, via embedded computer-based "intelligence," are assuming certain humanlike characteristics.

As industrial engineers, we are particularly interested in human interfaces with the work environment, machines, information/communication system, organizational/supervisory structure, and robot/"intelligent" machines. Each of these will be explored briefly in following sections.

#### *11.4.1 Human Interface with the Work Environment*

Industrial engineers design workplaces. These may be some combination of tables, desks, seats, chairs, and so on. Machines, equipment, telephones, computer terminals, and other devices are incorporated into the workplace according to the needs of the job.

Most workplaces are such that several different people will work there over a time interval. If the company is working multiple shifts, two or three different people may use the same workplace in one day. These factors must be considered when the I.E. specifies work-space dimensions. Where possible, adjustable chair heights and other dimensions are specified.

Considerable effort is also required to arrange various components of the workplace. Consider the workplace of an airline pilot. Where is the optimal location of the dozens of dials, gauges, levers, and buttons? Some of the components that must be arranged are visual displays, hand controls, foot controls, audio inputs, and audio outputs.

How much light (illumination) is needed for specific jobs? Considerable data exist that provide illumination standards for a variety of jobs. The widespread use of computer terminals has required modification of many lighting systems to reduce glare on the screens. Safety relative to lengthy exposure to CRTs is a growing concern.

Atmospheric conditions are also important in designing work environments. Temperature, humidity, and air cleanliness are the factors considered most frequently. If, for example, dust cannot be controlled adequately in the work environment, respirators should be furnished to all employees affected. Noise is still another atmospheric factor that must be considered when designing work environments. Excessive noise is not only annoying and detrimental to good work, but can be dangerous to the health of the employees.

Finally, the work environment should be safe and attractive. It should be

designed with good maintenance in mind. Good housekeeping practices are essential. Guards, carefully marked walkways, warning lights, and so on, are all important in designing work areas with employee safety in mind.

### 11.4.2 Human Interface with Machines

Virtually all human beings interface with machines of one type or another. The human/machine interface is obvious for truck drivers, lathe operators, crane operators, typists, and computer operators. In most productive work systems involving people and machines, the human being serves a control function. The person adjusts the workpiece in the machine, pushes the start button, watches as the machine operates, stops the machine in emergencies, decides when to shut the machine off, and so on.

Human factors data are extremely helpful in selecting the size, shape, and location of various controls. Consider the human-machine interface involved in driving an automobile. All the displays need to be designed carefully to be immediately recognizable. They also need to be located such that they are easily visible, even when the auto is in motion. The steering wheel needs to be correctly sized so that the driver has the proper "feel" of the road as he makes steering corrections. The brake pedal must be located properly and the correct amount of resistance built in. Warning lights and signals indicate "out-of-norm" conditions. If the oil in the crankcase is too low, a red light comes on. If the cooling system loses its coolant, the temperature light comes on. Recent automobiles provide verbal warnings.

One of the most important tasks of the I.E. is to allocate functions among machines and human beings in the system. In making this allocation, some general guidelines can serve to delineate between people and machines.

In general, human beings are better in performing the following types of tasks:

- (1) Recognizing complex patterns, which may vary considerably over time.
- (2) Recognizing unusual and unexpected events.
- (3) Storing large amounts of information over long time periods, then assimilating this information into general principles and strategies. (This is the basis of intuition.)
- (4) Quickly retrieving pertinent information; using associative abilities to retrieve many related items of information and to project these to inferences involving complex associations between related information.
- (5) Adapting the decision process to differing situations.
- (6) Reasoning inductively, generalizing from specific observations.
- (7) Making subjective evaluations and decisions.
- (8) Discerning the most important activities to concentrate on when overload exists.

In general, machines are better in performing the following types of tasks:

- (1) Monitoring for prespecified events.
- (2) Storing quantitative data quickly and in great volume.



- (3) Retrieving and processing quantitative data quickly.
- (4) Performing repetitive activities reliably over extended periods of time.
- (5) Responding to inputs rapidly and consistently.
- (6) Ignoring distractions.

These and other factors are considered when allocating functions among machines and human beings in a complex work system.

#### ***11.4.3 Human Interface with Information/Communication Systems***

The human being is an essential component in essentially all information/communication systems. These systems are becoming very complex in most modern organizations.

People receive information person to person and from written reports, information displays, telephone calls, buzzers, lights, signals, dials, sounds, and so on. People transmit information in writing, through keyboards, verbally, and by actuating certain input devices.

This is a complex subject. For our purposes, we will examine one small portion of this topic by considering when to use auditory versus visual presentation of information. This is summarized in Figure 11.2.

#### ***11.4.4 Human Interface with Organizational/Supervisory Structure***

This particular interface has to do with the entire set of issues involving job design, reporting relationships, performance appraisal, evaluation, and training. Details of many of these issues are covered in Chapter 21. These issues are also closely interwoven with the concepts discussed in Section 11.3.

| <i>Use Auditory Presentation if:</i>                                                                                                                                                                                                                                                                                                                                                                        | <i>Use Visual Presentation if:</i>                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>● Message is simple</li> <li>● Message is short</li> <li>● Message not referred to later</li> <li>● Message deals with events in time</li> <li>● Message calls for immediate action</li> <li>● Visual systems of the person is overburdened</li> <li>● Receiving location is too bright</li> <li>● Person's job requires worker to move about continually</li> </ul> | <ul style="list-style-type: none"> <li>● Message is complex</li> <li>● Message is long</li> <li>● Message will be referred to later</li> <li>● Message deals with location in space</li> <li>● Message does not call for immediate action</li> <li>● Auditory system of the person is overburdened</li> <li>● Receiving location is too noisy</li> <li>● Person's job allows worker to remain in one position</li> </ul> |

**Fig. 11.2.** When to use auditory or visual presentation [From B. H. Deatherage, "Auditory and Other Sensory Forms of Information Presentation," in H. P. Van Cott and R. G. Kinkade (eds.), *Human Engineering Guide to Equipment Design*, rev. ed. (Washington, D.C., U.S. Government Printing Office, 1972).]



We made the point in Section 11.3 that most people respond positively when they know the objectives of their organization, what their own jobs are, and that they will be treated fairly, consistently, and with respect. People want to be treated as individuals and not as numbers or as blocks on an organization chart. They also want to have some input into decisions that affect them directly, such as the arrangement of the workplace and the inspection of the work they perform.

During the early days of the American industrial revolution, (roughly 1850–1920), the work force in our factories came primarily from farms and from the masses of people who were immigrating to the United States. These people were largely uneducated and unskilled. As the industries grew from small numbers of workers (say 30) to very large work forces (several thousand) the company managers were faced with the problem of how to group (organize) the workers for maximum efficiency. The early efforts at organizing were largely trial and error.

Organizing and managing large work forces was a new endeavor. The only large organizations in the past had been military organizations. Consequently, company managers adopted many of the practices of military organizations, this being the only experience base from which to draw.

The result has been the rigid, formal-looking organization charts that we see on company walls that are supposed to represent authority/responsibility relationships. We hear a lot about “the chain of command,” or “we must go through channels.” Yet that is not how real organizations actually work.

The military model also gave managers their supervisory style, which basically amounts to “do what you are told, don’t ask questions, and have unswerving loyalty to the company.” Unfortunately, this is the way the supervisory style actually works in many organizations.

There are no clear-cut solutions to these problems at the present time, but we are beginning to see some alternative styles of organization and supervision. These are largely experimental, and we have a long way to go.

A great amount has been written in recent years about “quality circles,” “Japanese management style,” “participatory management,” and so on. The industrial engineer must play a central role in finding more effective means of organizing and supervising the work force of organizations.

#### ***11.4.5 Human Interface with Robots and Intelligent Machines***

If any one thing symbolizes the second industrial revolution that is now under way, it is the robot. Although robots are certainly an important aspect of the new manufacturing technology, they are by no means the dominant driving force.

Robots and robotic devices have been sensationalized by the popular press and by science fiction movies. In addition, there is a concern among many blue-collar workers that robots will displace them. Thus, to many uninformed people, robots are mysterious things that appear to be threatening and to have humanlike qualities.

One thing we will *not* see is a bunch of two-legged robots marching into a

factory having today's typical equipment, and operating the system in the same manner as would human operators. In those situations where robots can play an important role, the total set of manufacturing equipment with which the robot must interact is specifically designed with the robot in mind. It would be foolish (or at best poor design) to consider robots as one-to-one replacements of human beings with all other equipment unchanged. Such an approach would not take advantage of the robot's superiority to the human being in certain areas, nor would it consider the inferiority of the robot in other areas.

Many early applications of robots did not take advantage of their capabilities, leading one manager to remark that his industrial engineers should design a job enrichment program for the robot! In this particular case, a large, expensive robot was simply grasping items coming from a conveyor, raising them 2 feet, and then loading them onto a take-away conveyor that ran at right angles to the first conveyor. The robot possessed accuracy and repeatability capabilities on the order of 10 times greater than were needed. Furthermore, a fairly standard transfer mechanism using hard automation concepts would have been far superior and much less expensive.

Many factories of the future will contain numerous robots. Within the foreseeable future, there will also be many human beings in these factories. Industrial engineers must develop methods for designing productive work systems which include a varied mix of human beings, robots, and other "intelligent" machines.<sup>2</sup>

There are at least three sets of issues involving the interface between human beings and intelligent machines. The first set of issues is concerned with how to allocate work functions and control functions optimally among the human and machine elements comprising the system. Figure 11.1 compared certain performance characteristics of human beings and "unintelligent" machines. A comparable analysis would have to be conducted for each different type of intelligent machine.

The second set of issues concerning the human being and intelligent machine interface is that of safety. A robot or any machine will activate its components (such as swinging an arm) in response to specific commands. It will not know to stop its movement if a human being happens to be in its path. There have been several confirmed deaths of human beings caused by robot arm movement. Considerable effort is being directed toward the provision of safety features on robotic systems. However, this is expected to remain a critical issue for many years to come.

The third set of issues concerning the interface between human beings and intelligent machines has to do with the question of "who has the final say" when an intelligent machine wants to do one thing and the person wants to do another. At what point, and under what conditions, should a human being be permitted to override a decision made by an intelligent machine? What about the reverse situation? Should a machine ever be permitted to override the decision of a human being?

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<sup>2</sup>The term *machine intelligence* is frequently used to characterize such capabilities as machine vision (image processing), pattern recognition, audio input/output, tactile sensing, and adaptive control (learning, or modifying its own behavior).

Research into these issues will be pursued for many years. Industrial and systems engineers have an opportunity to contribute to this important effort.